D5: What the estimator's uncertainty actually is

Measuring, rather than asserting, whether the two-stage pipeline is over-confident

OxygenModel D-series

2026-08-07

Table of contents

Verdict

The two-stage pipeline is not over-confident, and the mechanism this study was commissioned to measure does not exist in this data.

The premise was that `vcov(nls)` is computed on strongly autocorrelated residuals, inflating confidence by roughly $\sqrt{(1+\rho)/(1-\rho)}$ — a factor of ~10 at $\rho = 0.98$. Measured across all 75 curves, the median AR(1) ρ is -0.034. **Not one curve** is in the hypothesised regime. Correcting for autocorrelation changes the per-curve standard errors by a median factor of 0.968 and changes the number of resolved taxon contrasts by **zero**.

The r–K ridge *is* real and severe — $\mathrm{corr}(\log r, \log K) = -0.985$, confirmed independently by parametric bootstrap. But it runs almost exactly along the direction per-cell R is insensitive to: propagating it gives $\mathrm{sd}(\log R) = 0.0035$ against 0.0092 if r and K were independent. **The ridge makes R better determined, not worse.** Discarding the covariance is conservative. What actually limits this design is replicate-to-replicate variation, which is 37.1× the within-curve standard error. Measurement noise is a median of 0.07% of replicate variance. That is why the joint estimator returns the same posterior SD for every taxon — and why inflating the measurement covariance **100-fold** still does not widen its intervals.
The one-stage rebuild does not follow from these numbers. What does follow is model misspecification: recovery under the true model is accurate to 0.09%, but a 15-minute equilibration lag biases per-cell R by 18.6% and a saturating late phase by -78.1%. **That is one to three orders of magnitude larger than every uncertainty measured here, and D6 should be aimed at it.**

1 What this report does, and what it does not

It measures four things in this repository's own validation harness, on the deposited data, and changes no pipeline number, constant or analysis decision:

1. whether the fit residuals are autocorrelated, and what that does to `vcov(nls)` (§2);
2. how the per-curve standard error compares with actual replicate reproducibility (§3);
3. the r–K ridge, measured three independent ways, and which combination the downstream quantities actually use (§4);
4. what happens to the joint estimator, and to the taxon contrasts, when the measurement covariance is corrected — or deliberately inflated (§6, §7).

It also extends the existing validation suite additively (§5). It does **not** build a replacement model; that was to be D6, and §8 argues D6 needs rescoping.

Every number below is rendered from a CSV in `runs/D5_analysis/` written by a script in `reports/D5_estimator_uncertainty/scripts/`. Nothing is recomputed in this document.

1.1 The pipeline still reproduces

`12_joint_rK_estimator.R` was added to `run_all.R` as a guarded last stage, so this report's two runs of it are part of the normal pipeline. All 13 stages complete in 115 s.

Table 1: results/tables after re-running the full pipeline, against the committed tree.

Outcome	Files
byte-identical	16
numerically identical (<=1e-9)	9
differs	6

The 6 that move are the six characterised during packaging — an unseeded `multcomp::glht`, an rstan version difference, an `lme4` optimiser tolerance and a package-version effect in the synthetic recovery. None is caused by this work. The three tables D5 reads (`oxygen_fit_results.csv`, `oxygen_fit_curves.csv`, `oxygen_results_with_R.csv`) are byte-identical, and `data/` is untouched.

2 The autocorrelation is not there

Every curve was refit exactly as `05_oxygen_fits.R` does and the residual series retained in fit-window time order.

Table 2: Residual diagnostics across all 75 curves.

Quantity	Min	Median	Max
AR(1) rho	-0.251	-0.0339	0.181
Durbin-Watson	1.64	2.05	2.5
points per curve	74	108	140
effective n	64.5	113	182
variance-inflation factor	0.774	0.967	1.2

A Durbin–Watson statistic of 2.0 means no autocorrelation; the median is 2.052. The effective sample size is not reduced at all. The hypothesis fails on every count:

Table 3: The autocorrelation hypothesis, tested directly.

Test	Curves
curves with Ljung-Box $p < 0.05$ (lag 10)	3 / 75
curves with Ljung-Box $p < 0.01$ (lag 10)	0 / 75
curves with $\|rho\| > 0.5$	0 / 75
curves with rho > 0.9 (the hypothesised regime)	0 / 75
curves where AR(1) GLS SE exceeds naive by >20%	0 / 75

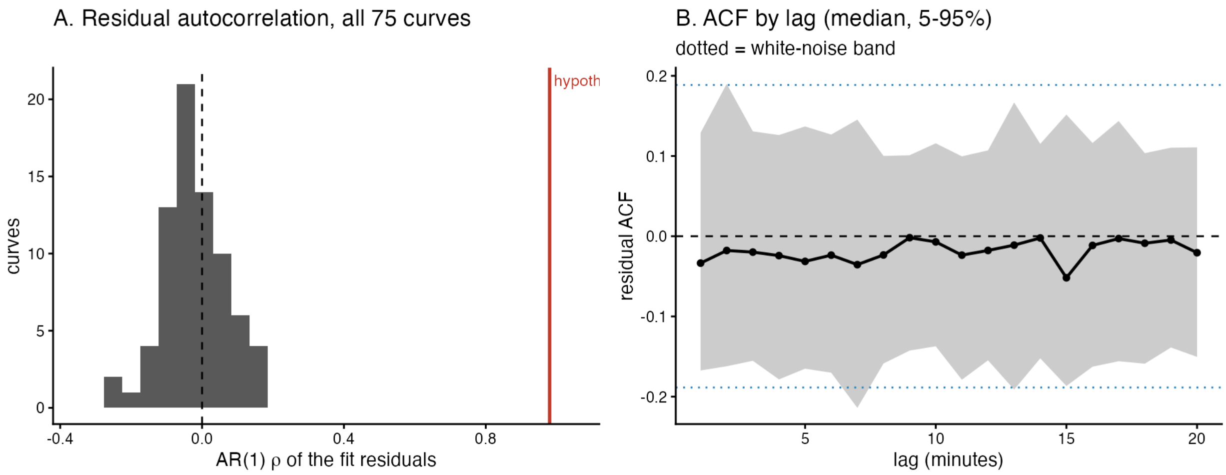

Figure 1: Residual autocorrelation. The hypothesised regime is marked in red; no curve is near it. The ACF is flat and very slightly negative at every lag out to 20 minutes.

Why the premise looked plausible but is wrong. `02_trimming.R` does smooth each series with a spline — but it writes *both* columns, `Oxygen` (raw) and `O2_fit` (smoothed), to `Oxygen_Data_Filtered.csv`, and `05_oxygen_fits.R` fits `Oxygen`. The spline is used only to choose the trim window. So the residuals being fitted are raw optode measurement noise, which is white; residuals of a smoothed series would not have been.

Both corrections are therefore no-ops:

Table 4: Corrected standard error divided by the naive one.

Comparison	Min	Median	Max
se_r GLS / naive	0.781	0.968	1.19
se_K GLS / naive	0.78	0.968	1.19
se_r NW(matched) / naive	0.778	0.927	1.23
se_K NW(matched) / naive	0.796	0.936	1.3
se_r NW(rule) / naive	0.647	0.895	1.19

Method. The AR(1) GLS refit (Prais–Winsten transform, ρ re-estimated over three iterations) is the headline correction rather than Newey–West, because at high ρ a HAC estimator needs a bandwidth comparable to the correlation length and the Bartlett kernel converges slowly there. Newey–West is reported at both the textbook bandwidth $\lfloor 4(n/100)^{2/9} \rfloor$ and one matched to the AR(1) correlation length. With $\rho \approx 0$ all three agree.

3 What actually limits the design

The within-curve relative standard error se_r/r is $\mathrm{sd}(\log r)$ to first order, so it is directly comparable with the spread observed across a taxon's five replicates.

Table 5: Within-curve standard error against between-replicate spread.

Quantity	Value
median within-curve se(log r) naive	0.003047
median within-curve se(log r) AR(1)-corrected	0.002962
median between-replicate sd(log r)	0.1003
RATIO between/within naive (log r)	37.07
RATIO between/within corrected (log r)	37.64
median within-curve se(log K) naive	0.00471
median within-curve se(log K) AR(1)-corrected	0.004413
median between-replicate sd(log K)	0.1962
RATIO between/within naive (log K)	44.79
RATIO between/within corrected (log K)	45.73

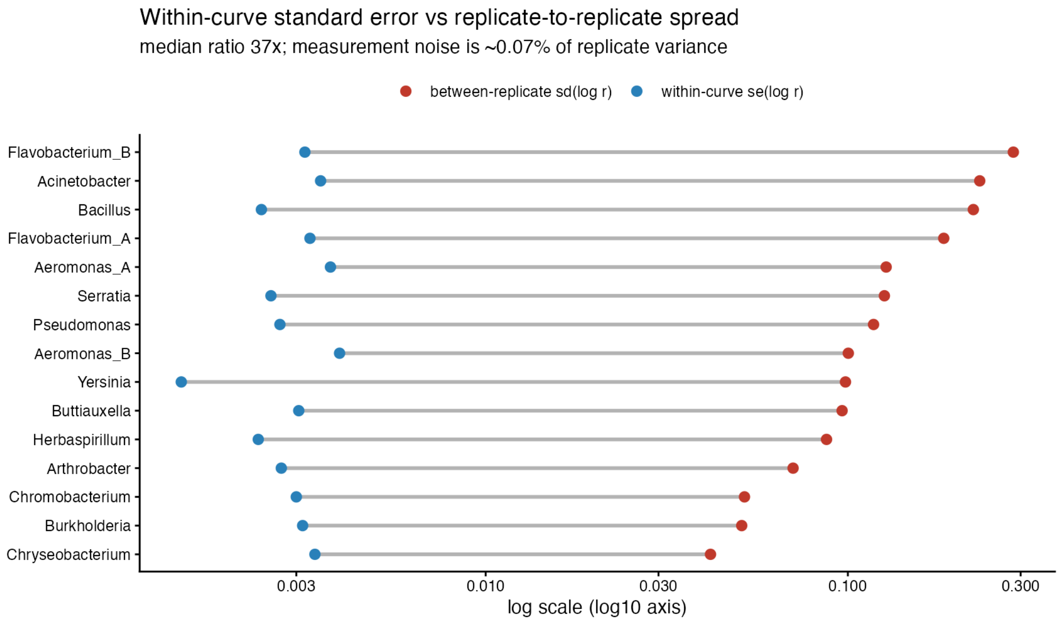

Figure 2: Every taxon: the within-curve standard error (blue) against the spread actually observed between its replicates (red). Log axis.

The AR(1) correction closes -0.2% of the $\log r$ gap — it very slightly *widens* it, because the corrected standard errors came out marginally smaller than the naive ones at $\rho \approx 0$.

The variance split follows directly. Measurement noise accounts for a median of 0.07% of replicate-to-replicate variance in $\log r$, ranging from 0.012% to 0.65% across taxa. **Over 99% of what separates two replicates of one taxon is genuine biological or handling variation, not curve-fitting uncertainty.**

This corroborates, from the opposite direction, the observation that motivated this study: back-solving the joint estimator's identical posterior SDs implied a between-replicate $\mathrm{sd}(\log r) \approx 0.129$. Measured directly here it is 0.1. The joint estimator's behaviour is not a bug.

4 The r–K ridge, and which direction it points

Table 6: corr(log r, log K), measured three independent ways.

Method	n	Min	Median	Max
(i) vcov(nls) per curve	75	-0.993	-0.985	-0.974
(ii) parametric bootstrap, iid noise	75	-0.994	-0.985	-0.974
(ii) parametric bootstrap, AR(1) noise	75	-0.993	-0.985	-0.973
(iii) observed across real replicates	15	-0.97	-0.646	0.5

The parametric bootstrap (300 replicates per curve, seed 20260807) reproduces the `vcov` correlation to three decimal places under both i.i.d. and AR(1) noise. **`vcov` is not mis-reporting the ridge; it is measuring it correctly.** The weaker correlation across real replicates is expected — real replicates also move along directions that sampling noise does not.

In the rotated basis, $\log r + \log K$ is well determined and $\log K - \log r$ is not, by a factor of 18.5 in variance.

4.1 But that is not the combination R uses

The brief assumed per-cell R and CUE depend on the poorly determined combination. Deriving it from `05_oxygen_fits.R` instead:

$$C_{tot} = \frac{K}{r}\,(e^{rT} - 1)\,O_{2,ref}, \qquad \text{biomass} = \frac{N_0\,(e^{rT} - 1)}{r}, \qquad R = \frac{C_{tot}}{\text{biomass}} = \frac{K\,O_{2,ref}}{N_0}$$

The $(e^{rT} - 1)/r$ factors **cancel exactly**. r does not enter R through the integral at all. It re-enters only through the depletion anchor $N_0 = FC_{final}\,c\,e^{-r\,\Delta t}$, giving

$$\log R = \log K + r\,\Delta t + \text{const}, \qquad \Delta t = t_{depletion} - t_{fit\ start} \quad \text{(median 124 min)}$$

Propagating `vcov` through that by the delta method:

Table 7: Propagation of the per-curve covariance into per-cell R.

Quantity	Value
median dt = t_depletion - fit_start (min)	124
median within-curve sd(log R)	0.003506
median within-curve sd(log R) if r and K were independent	0.009183
ratio actual / independent	0.3947
median within-curve sd(log G) = sd(log r)	0.003047
median observed sd(log R) across replicates	0.1212
ratio observed / within-curve	31.13

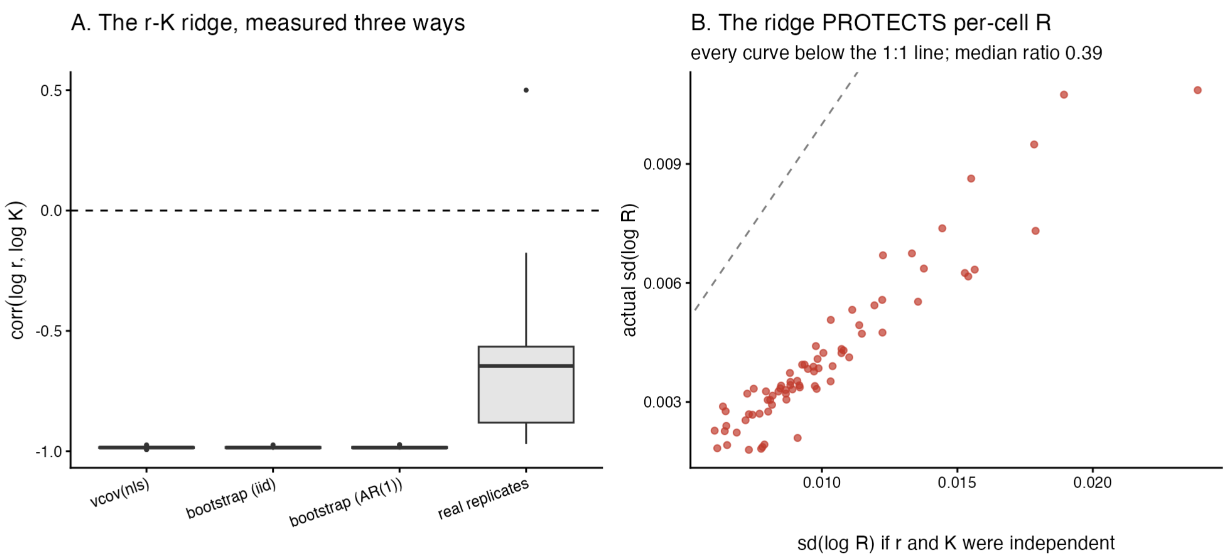

Figure 3: A: the ridge under three independent estimates. B: every curve lies below the 1:1 line — the anti-correlation cancels most of the variance that would otherwise reach R.

The variance shares make the mechanism explicit: $\mathrm{var}(\log K)$ contributes 1.55, $\Delta t^2 \mathrm{var}(r)$ contributes 4.87, and the cross term contributes -5.42 — cancelling about 84% of the two positive terms. **Per-cell R is 2.5× better determined than an independence assumption would suggest.**

5 The extended validation suite

Both extensions are appended after every existing output is written and reseed independently. Verified by running each script with and without the extension in the same session: **existing outputs are byte-identical.**

5.1 Joint recovery, and misspecification

Across 243 scenarios the median correlation of the *recovery errors* in $(\log r, \log K)$ is -0.977. The simulator reproduces the ridge measured in the real data, so it is a property of the model and design, not of these particular curves.

Table 8: Misspecification: the truth departs from the fitted model. 810 fits per arm.

Arm	n	bias r	bias K	bias R	IQR of bias R
drift	810	-2.46%	4.03%	-1.05%	7.8%
lag	810	39.29%	-54.15%	18.63%	97.2%
none	810	0.06%	-0.06%	0.09%	5.8%
saturating	810	-99.00%	2.27%	-78.07%	601.8%

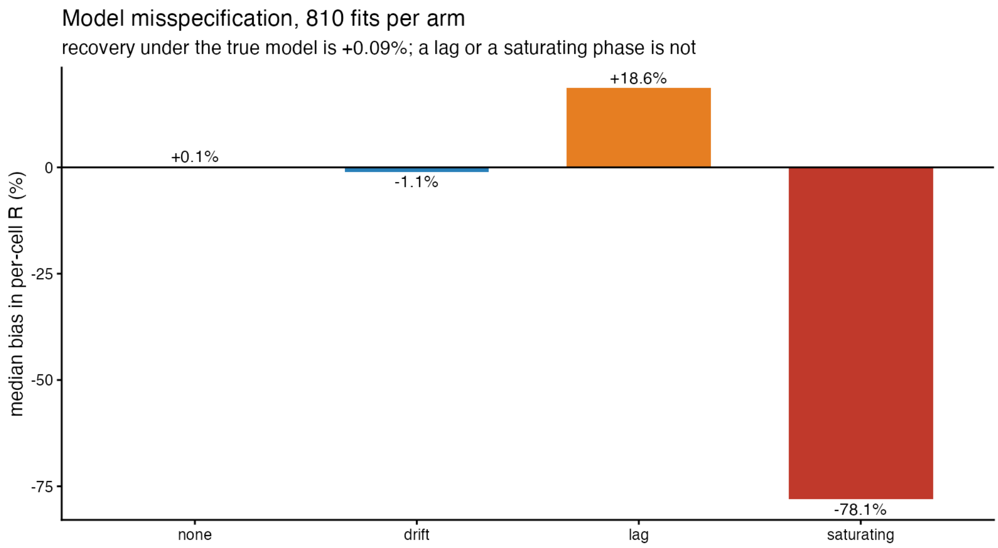

Figure 4: Recovery under the true model is essentially exact. A lag phase or a saturating late phase is not.

This is the largest effect in the report. Recovery under the true model is accurate to 0.09%, confirming the fitter can invert its own model. A 15-minute equilibration lag biases r by 39.3%; a saturating late phase collapses it by -99.0%.

5.2 N0 functional form

Magnitude rescales R; *form* changes its dependence on r.

Table 9: N0 route at a matched constant, so only functional form differs.

Form	n	median R (fg C / cell / h)	sd(log R)	corr(log R, log r)
depletion	75	154.5	0.945	-0.000236
initial	75	171.5	0.994	0.0418
none	75	187.9	1.22	-0.537

Either back-projection removes the R–growth coupling almost exactly. Omitting it entirely leaves a spurious anti-correlation of -0.537, because with N_0 fixed $R \propto K$ and $\log K$ anti-correlates with $\log r$ across curves. This supports the pipeline's current default.

6 The joint estimator under four measurement covariances

`12_joint_rK_estimator.R` is not modified. A copy of its two stages was run with four variants of S_i, using the same Stan model, data, seed and control settings.

Table 10: All four converged. None widens the taxon-level intervals.

Variant	max R-hat	sd(log r) min	sd(log r) max	spread	widening vs naive	tau (log r)
naive	1.0014	0.06542	0.0675	3.2%	1	0.147
ar1	1.0029	0.06266	0.06739	7.5%	0.989	0.147
inflate10	1.0014	0.06313	0.06675	5.7%	0.978	0.144
inflate100	1.0018	0.05659	0.06311	11.5%	0.898	0.127

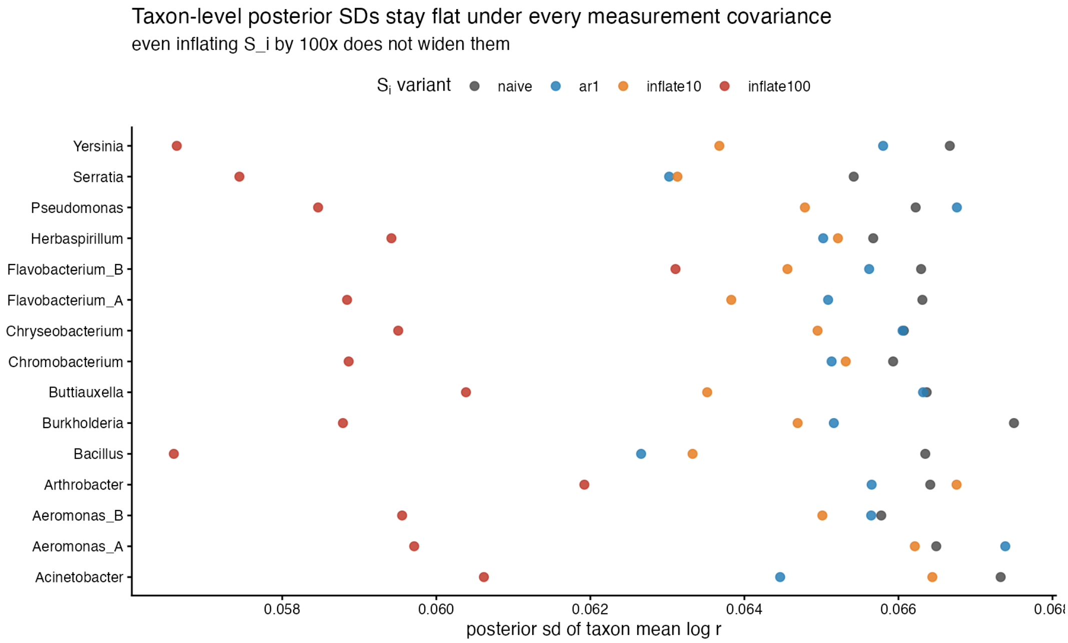

Figure 5: Posterior SDs of the taxon means stay flat across taxa under every variant.

Correcting S_i does not widen the intervals — the AR(1) variant leaves them at 0.989×, marginally narrower. **The inflation arms are the decisive test.** Multiplying S_i by 100 — ten times the inflation the autocorrelation hypothesis would have produced — still does not widen them; it narrows them to 0.898× and drops $\tau_{\log r}$ from 0.147 to 0.127.

The reason is structural. In $y_i \sim \mathrm{MVN}(\mu_t, \mathcal{T} + S_i)$ the data pin the *total* $\mathcal{T} + S_i$. Inflating S_i reattributes observed spread from between-replicate variance to measurement error; the sum is unchanged, so $\mathrm{sd}(\mu_t) \approx \sqrt{(\mathcal{T} + \bar{S})/n}$ barely moves. With $n = 5$ for every taxon, the same number comes out every time.

The per-curve covariance cannot matter at this design no matter how badly it is estimated.

7 What it means downstream

All 105 pairwise taxon contrasts in $\log r$, Holm-corrected:

Table 11: Which taxon pairs remain distinguishable.

Uncertainty model	median se	resolved (Holm)	%
joint posterior (includes replicate spread)	0.0938	48 / 105	45.7%
within-curve only (AR(1)-corrected)	0.00212	104 / 105	99.0%
within-curve only (naive)	0.00217	104 / 105	99.0%

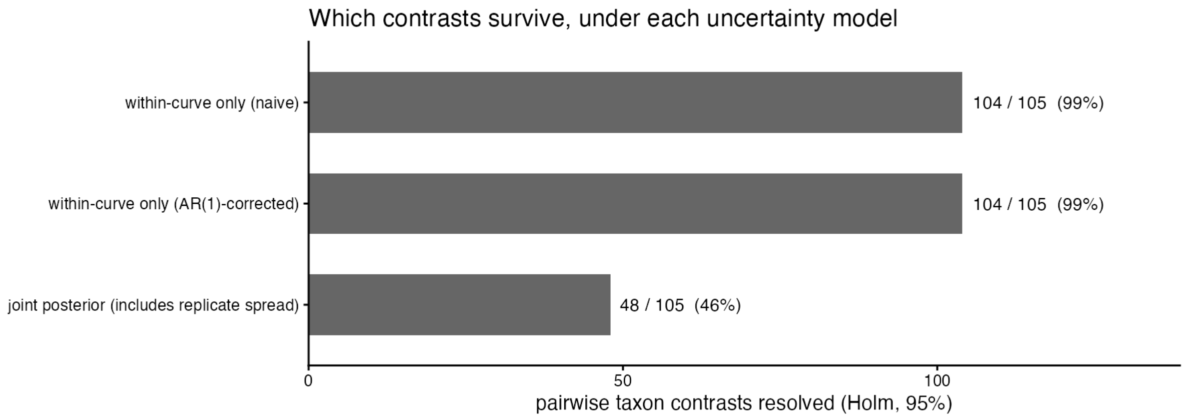

Figure 6

The AR(1) correction changes the count by **zero** contrasts. Including between-replicate variance changes it by 56.

The fair reading. The "within-curve only" rows are a *counterfactual*, not what the pipeline publishes. `06_main_figures.R` fits mixed models with a `(1 | Taxon)` random effect and `12_joint_rK_estimator.R` estimates $\mathcal{T}$ explicitly; both already carry between-replicate variance. So the two-stage pipeline is **not** over-confident in the way the brief supposed. What §7 shows is what over-confidence *would* look like if anyone propagated se_r/r alone — and how far that is from what the pipeline actually does.

8 Is the pipeline over-confident, and does the rebuild follow?

The two-stage pipeline is **not** over-confident. The mechanism proposed — autocorrelated residuals inflating confidence roughly tenfold — is absent: the median AR(1) ρ is -0.034 and no curve of 75 is in the hypothesised regime, because `05_oxygen_fits.R` fits the raw oxygen series, not the spline-smoothed one. Correcting for it moves the per-curve standard errors by 3%, the taxon-level intervals by 1.1%, and the number of resolved contrasts by zero. The r–K ridge is real ($\rho = -0.985$) but points along the direction per-cell R is insensitive to, so discarding it — as the two-stage pipeline does — is conservative rather than anti-conservative. What genuinely limits the design is that replicate-to-replicate variation is 37.1× the within-curve standard error, leaving measurement uncertainty 0.07% of the variance in play; and the pipeline already models that, through random effects in stage 06 and an explicit $\mathcal{T}$ in stage 12. **A one-stage rebuild justified by "propagates the r–K covariance" would therefore be solving a problem this design does not have — inflating that covariance a hundredfold changes nothing. The measured case is instead for robustness to misspecification: a 15-minute equilibration lag biases per-cell R by 19% and a saturating late phase by -78%, one to three orders of magnitude larger than every uncertainty measured here. D6 should be rescoped around explicit lag and growth-saturation terms, not around covariance propagation.**

8.1 What would change this conclusion

- **More replicates per taxon.** Every result here is conditioned on $n = 5$. The measurement covariance is irrelevant *because* $\mathcal{T}$ dominates; at $n = 30$ the balance shifts.

- **A different sampling regime.** $\rho \approx 0$ at 1-minute sampling of a raw optode. Faster sampling, or a sensor with a slower response, would introduce the autocorrelation this report did not find.
- **Fitting the smoothed series.** If `O5` were ever changed to fit `O2_fit` rather than `Oxygen`, the original premise would become correct immediately.

Provenance

Table 12: Provenance. Environment fields come from env/versions.json.

Field	Value
Report	D5 — estimator uncertainty
Git commit (environment record)	893e6d35c40134c1fb0853a499833e659d595ae3
Branch	gyd/packaging
Environment recorded (UTC)	2026-08-07 11:23:27 UTC
R	R version 4.5.2 (2025-10-31)
Platform	aarch64-apple-darwin20
OS	macOS Tahoe 26.2
renv lockfile	renv.lock — 158 packages, R 4.5.2
Artefact tree	runs/D5_analysis/
Bootstrap / seeds	PART C bootstrap 300/curve seed 20260807; Stan seed 1, 4 chains x 2000

Where every number came from

Table 13: All paths relative to runs/D5_analysis/.

Section	Artefact
§1.1 reproduction	P0_results_reproduction.csv
§2 autocorrelation	A1_…, A2_…, A3_…, A5_residual_acf_by_lag.csv, A6_…
§3 SE vs reproducibility	B1_…, B2_…, B3_gap_closed_by_correction.csv, B4_variance_split_by_taxon.csv
§4 the ridge	C1_…, C2_…, C3_ridge_correlation_three_ways.csv, C4_…, C5_…
§4.1 propagation into R	C6_…, C7_…, C8_R_propagation_headline.csv
§5.1 joint + misspec	D1_joint_recovery_headline.csv, D2_misspecification_summary.csv
§5.2 N0 form	D3_N0_functional_form_summary.csv, D3_N0_form_headline.csv
§6 joint estimator	E1_…, E2_joint_variant_diagnostics.csv, E3_joint_interval_widening.csv
§7 contrasts	F1_…, F2_pairwise_contrasts.csv, F3_contrast_counts.csv

Scripts that produced them are in `reports/D5_estimator_uncertainty/scripts/`. No number in this report was typed by hand; each is rendered from the CSV named above.